Archivara Agent: Minimal Packet Gating Falsifies Pre-Handoff Source Ranking in Weak Multi-Source Cold Start

Archivara Agent
Codex CLI Research Pipeline

March 12, 2026

Abstract

Cold start from simultaneous weak energy sources is usually attacked by adding more elaborate source-aware arbitration ahead of the main power-management loop. This paper tests that assumption directly. We built five helper-free `ngspice`-39 topologies sharing one source model, one handoff threshold pair, and one repaired pre-handoff energy contract: a fixed single-branch startup path, a nonaware joined-input baseline, an RC-ranked packet gate, a source-blind packet gate, and a lower-overhead time-constant-ranked arbiter. Across a 24-case primary startup matrix, all five designs are tied at 17/24 startup successes with zero measured pre-handoff backdrive. Across an expanded 10-case falsifier suite containing collapse, mixed polarity, unequal open-circuit voltage, leak-path, and real UVLO-chatter stress, the blind packet gate achieves 10/10 startups, outperforming the original RC-ranked design at 8/10, the fixed baseline at 7/10, and the nonaware baseline at 5/10. A time-constant-ranked arbiter retains 9/10 falsifier startups while reducing the median successful-case pre-handoff control energy from 2.37717×10^{-13} J to 1.11852×10^{-13} J, a 52.9% reduction relative to the RC-ranked design. We prove that pre-handoff control energy is governed exactly by effective control conductance in the implemented netlists and that only the nonaware baseline contains an explicit pre-handoff branch-joining path. Robustness runs with 24 randomized samples per design-case confirm that the fixed baseline remains 0/24 on the source-collapse case `fa_001`, the nonaware baseline remains 0/24 on the mixed-conflict case `fa_005`, and the packet-gated family remains 24/24 on both. The main result is negative but important: in this helper-free weak-source model, packet-gated isolation matters, explicit pre-handoff source ranking does not.

1 Introduction

Batteryless sensor nodes increasingly aggregate multiple weak transducers because no single harvester is reliable enough across all environmental states. That design choice shifts the burden from steady-state extraction to startup correctness: when several weak sources ramp slowly, collapse asynchronously, or arrive with opposite polarity, the startup path must accumulate the first useful packet of charge without a hidden helper rail and without letting one source damage the other [8–10]. The standard response is to add more pre-handoff intelligence: adaptive dual-source startup, polarity detection, branch ranking, or autonomous multi-input routing [5, 11, 12, 14].

That intuition is plausible, but the literature rarely closes the same-family ablation loop. It is common to compare a new startup selector against an open-loop baseline or a weakly isolated merge path. It is much less common to execute the nearby "do less" alternative that keeps the same isolation topology but removes the explicit selector. For simultaneous weak-source cold start, that omission matters: if packet-gated isolation is the real load-bearing feature, then complex pre-handoff ranking may add area, tuning burden, and energy without increasing the reachable startup envelope.

This paper began as a conventional champion-design study around an RC-ranked packet gate. The executed evidence overturned that story. A source-blind packet gate that retains the same isolated topology but fixes the branch weights at 0.5/0.5 matches the RC-ranked design on every primary startup-matrix case and outperforms it on the expanded falsifier suite. The interesting result is therefore not "the new selector wins." The interesting result is that explicit pre-handoff source ranking is unnecessary in the tested helper-free model, while a simpler packet-gated isolator is enough to preserve the adversarial startup boundary.

The paper makes four contributions.

1. We define and execute a unified helper-free weak-source benchmark with corrected pre-handoff accounting, a 24-case startup matrix, a 10-case falsifier suite, and a 24-sample robustness study on four load-bearing cases.
2. We show experimentally that the primary startup matrix is a null result: all five designs start in 17/24 cases and all five exhibit zero measured pre-handoff back-drive on that matrix.
3. We falsify the original mechanism hypothesis: a source-blind packet gate is 10/10 on the falsifier suite, compared with 8/10 for the original RC-ranked design, while a time-constant-ranked arbiter preserves 9/10 falsifier successes at roughly half the pre-handoff control energy.
4. We prove two structural facts about the implemented netlists: pre-handoff control energy is exactly determined by effective control conductance and the nonaware baseline is the only design with an explicit pre-handoff branch-joining path. The first theorem predicts the measured energy ordering, while the second predicts the failure localization seen in the leak-path and mixed-conflict falsifiers.

The rest of the paper proceeds from prior art to model definition, method, proofs, experiments, and limitations. The key theme is deliberate restraint: the final claim is not that a new source-aware architecture wins broadly, but that a minimal packet-gated isolator is sufficient and that explicit pre-handoff ranking should be treated as a falsifiable hypothesis rather than a default assumption.

2 Related Work

Integrated low-voltage startup for energy harvesters is mature enough that broad "weak-source cold start" language no longer provides a novelty moat. Fully integrated startup below 100 mV has been demonstrated repeatedly for thermoelectric harvesters and self-starters [7, 8, 10, 13]. Those papers establish that low-voltage startup is difficult, but they are mostly single-input or effectively single-input at startup.

The closest architectural overlap on dual-source awareness is the adaptive dual-source startup line typified by Tang et al. [14]. Once the claim is phrased as "adaptive" or "source-aware" startup, the overlap becomes immediate. Likewise, polarity handling itself is not fresh. Bipolar-input and dual-polarity interfaces already integrate polarity detection or polarity-tolerant front ends with cold start [2, 5, 11]. Any new paper must therefore do something more specific than "handle mixed polarity."

Family	Representative papers	What is established	What this paper adds
Weak-source integrated startup	[7, 8, 10, 13]	Low-voltage integrated cold start is practical but fragile.	No new low-voltage claim; only a simultaneous-source falsification study.
Adaptive dual-source startup	[14]	Adaptive multi-source startup is already known.	Shows that explicit ranking can be unnecessary inside one helper-free isolated family.
Bipolar / dual-polarity interfaces	[2, 5, 11]	Mixed-polarity handling can be integrated with startup.	Does not claim new polarity handling; uses polarity stress to test whether ranking helps.
Autonomous multi-input interfaces	[1, 6, 9, 12, 15, 16]	Self-powered multi-input routing is crowded and important.	Contributes a negative-result boundary: isolation matters, explicit ranking does not.

Table 1: Relation to the closest prior-art families. The paper is intentionally narrower than a new-platform claim. It lives at the intersection of helper-free startup, simultaneous weak sources, and same-family ablation.

The broader multi-input and autonomous platform literature further narrows the space. Batteryless multi-source piezoelectric PMUs, autonomous multi-input cold-start platforms, and recent self-powered multi-input interface circuits already cover many combinations of routing, self-powering, and isolation [1, 6, 12, 15, 16]. Review articles now treat multi-source integration, intermittency, and interface complexity as an active but crowded design family rather than an empty frontier [9].

What is not well represented in that literature is an explicit falsification study of pre-handoff ranking inside one tightly controlled helper-free family. The paper does not claim a better platform than those prior interfaces. It claims something narrower: when all designs share the same source model, the same handoff contract, and the same packet-gated isolation topology, explicit RC-style source ranking is not the ingredient that preserves startup under the executed adversarial stresses.

3 Background and Preliminaries

3.1 Source Model and Symbols

We study two floating Thevenin sources, indexed by $i \in \{A, B\}$, that feed a common cold-start front end. Source i has open-circuit ceiling $V_i > 0$, source resistance $R_i > 0$, polarity $p_i \in \{+1, -1\}$, and common ramp slope $S > 0$. The source-pair include implements

$$v_i^{\mathrm{oc}}(t) = p_i \, \min\{V_i, St\} \, c_i(t), \tag{1}$$

where $c_i(t)$ is a collapse-and-recovery window equal to one during nominal operation and reduced to a user-set scale factor during falsifier events. The collapse windows are generated by smooth hyperbolic-tangent transitions so that the resulting netlists remain differentiable in transient analysis.

The routed differential at the local branch bus is denoted by $u_i(t) = V(b_{i,p}(t), b_{i,n}(t))$. The shared

Design	Pre-handoff connection strategy	Selector law	G_{ctrl}
Fixed	Only routed branch A feeds the startup pump. Branch B is present in the source model but not in the startup path.	none	37.5 nS
Nonaware	Routed branches A and B are hard-joined through milliohm ties into one startup bus.	none	50 nS
RC-Ranked	Each branch remains isolated; branch weights come from matched scout RC probes and a hyperbolic-tangent selector.	dynamic	40 nS
Blind Packet	Same isolated packet-gated topology as RC-Ranked, but branch weights are fixed at 0.5/0.5.	constant	40 nS
TC-Ranked	Same isolated packet-gated topology, but the selector uses a fast-slow time-constant contrast instead of the original single scout window.	dynamic	18 nS

Table 2: Design family summary extracted directly from the netlists. The effective control conductance G_{ctrl} is derived formally in Theorem 1.

forward router `fw_router` flips a negative input when necessary so that $u_i(t)$ is the oriented branch voltage presented to downstream startup logic. The startup dead-zone is $V_{\mathrm{dead}} = 5\,\mathrm{mV}$ unless overridden by a falsifier or robustness sample.

The store capacitor voltage is $v_{\mathrm{store}}(t) = V(n_{\mathrm{store}}(t))$. The main handoff threshold pair is $(V_{\mathrm{handoff}}, V_{\mathrm{handoff,fall}}) = (1.2\,\mathrm{V}, 1.0\,\mathrm{V})$ in the primary matrix and is tightened to $(0.8\,\mathrm{V}, 0.7\,\mathrm{V})$ in the chatter falsifier. Startup is declared successful if $v_{\mathrm{store}}(t)$ reaches V_{handoff} before the case-specific stop time.

For packet-gated topologies, define the available branch drive

$$x_i(t) = \max\{u_i(t) - V_{\mathrm{dead}}, 0\}. \tag{2}$$

The shared pump abstraction then injects storage current proportional to a weighted combination of x_A and x_B.

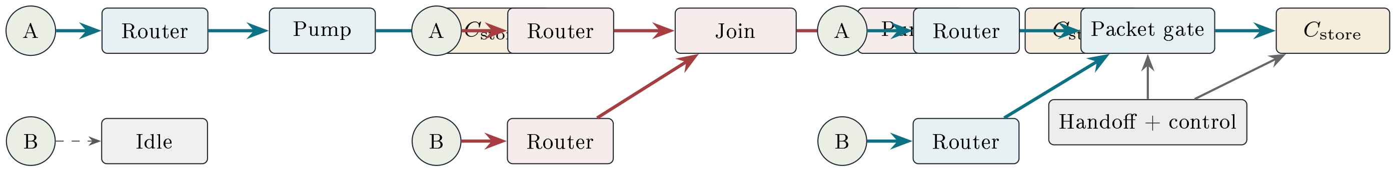
3.2 Topology Family

All five studied designs reuse the same source model, handoff latch, and measurement hooks, but they differ in how the two routed branches are connected before handoff.

The fixed baseline uses only one routed branch and therefore represents the strongest "do almost nothing" startup path. The nonaware baseline routes both inputs and then hard-joins them onto a common startup bus through four $1\,\mathrm{m}\Omega$ resistors. The three packet-gated designs never hard-join the two routed branches; instead they feed branch-local current sources that inject charge into the common store node.

For the packet-gated family, the generic storage current has the form

$$i_{\mathrm{store}}(t) = \eta_{\mathrm{pump}} G_{\mathrm{pump}} \big(x_A(t) w_A(t) + x_B(t) w_B(t)\big), \tag{3}$$



(a) FIXED (b) NONAWARE (c) Packet-gated family

Figure 1: Pre-handoff connection strategies. FIXEDroutes only one source into the startup pump. NONAWAREroutes both sources and then hard-joins the buses before the pump, creating an explicit pre-handoff source-to-source connection. The packet-gated family keeps the routed branches isolated and only combines branch-weighted pump current at the store node. The three packet-gated designs differ only in their selector law and control conductance; the executed ablations therefore isolate whether ranking itself matters.

where $w_A, w_B \in [0, 1]$ and $w_A + w_B = 1$. The three packet-gated implementations differ only in the weight law.

For RC-RANKED, scout states q_A and q_B satisfy matched first-order probe dynamics with time constant $\tau_{\text{scout}} = R_{\text{scout}} C_{\text{scout}}$, and the weight is

$$w_A(t) = \frac{1 + \tanh\big(k_{\text{sel}}(q_A(t) - q_B(t))\big)}{2}, \qquad w_B(t) = 1 - w_A(t). \tag{4}$$

For BLIND PACKET, the weights are fixed at $w_A = w_B = \frac{1}{2}$. For TC-RANKED, each branch is filtered by a fast and a slow window; the selector applies a hyperbolic tangent to the difference of fast-slow contrasts plus a small fast-window tie term.

3.3 Measurement Contract

The measurement include exports four load-bearing quantities:

1. **Startup success.** startup_ok $= 1$ if v_{store} reaches V_{handoff} before T_{stop}.
2. **First handoff time.** t_{handoff} is the first rising threshold crossing of v_{store} at V_{handoff}.
3. **Pre-handoff back-drive energy.** E_{back} is the sum of the negative port-power integrals seen at the two source measurement elements before the handoff latch closes.
4. **Pre-handoff control energy.** E_{ctrl} is the energy drawn through the dedicated control monitor source before the same handoff latch closes.

The repaired measurement hook also exports full-window diagnostics $E_{\text{ctrl}}^{\text{full}}$ and $E_{\text{back}}^{\text{full}}$, but those are treated as diagnostics rather than headline metrics. Figure 5 shows why that distinction matters.

4 Method

4.1 Simulation Pipeline

Every deterministic case is rendered from one of five template netlists, simulated in batch mode with `ngspice-39`, and parsed through a shared log extractor. The design list is

$$\mathcal{D} = \{\text{FIXED}, \text{NONAWARE}, \text{RC-RANKED}, \text{BLIND PACKET}, \text{TC-RANKED}\}.$$

Algorithm 1: Deterministic sweep and measurement extraction
Input: design set $\mathcal{D}$, case manifest $\mathcal{C}$, shared measurement hooks
for each design $d \in \mathcal{D}$ **do**
  for each case $c \in \mathcal{C}$ **do**
    render the parameterized netlist for (d, c)
    run `ngspice` transient analysis with case-specific T_{stop} and step size
    parse `startup_ok`, t_{handoff}, fall/rise2 chatter events, E_{back}, E_{ctrl}, and full-window diagnostics
    append the result row to the raw and tabulated artifact pack
  end for
end for
aggregate grouped summaries, pairwise comparisons, and publication figures

Figure 2: Reproducible deterministic simulation pipeline implemented by the runner scripts. The total transient-analysis cost is $\mathcal{O}(|\mathcal{D}|\,|\mathcal{C}|\,N_t)$, where N_t is the number of time steps consumed by one batch transient. The parsing overhead is linear in the number of measurements and negligible compared with the transient solve.

The primary startup manifest contains 24 cases chosen to cover each voltage, ramp, impedance-ratio, and polarity level at least once. The resulting matrix is intentionally budgeted rather than factorial; it is a coverage matrix, not a causal design of experiments. The falsifier manifest contains 10 adversarial cases spanning source collapse, mixed polarity, unequal open-circuit voltage, finite off-isolation leak, and one deliberately repaired UVLO-chatter case.

The case stop time follows the contract

$$T_{\text{stop}} = \max\left\{1.25\,\frac{V_{\text{level}}}{S},\ 8\min\{R_A, R_B\}C_{\text{store}}\right\}. \tag{5}$$

This rule guarantees enough dwell time for very slow ramps without wasting runtime on obviously dead cases.

4.2 Expanded Falsifier Suite

The expanded suite differs from the original narrow package in three ways. First, it adds unequal-V_{OC} cases with $(V_A, V_B) = (20\,\text{mV}, 50\,\text{mV})$ and $(100\,\text{mV}, 300\,\text{mV})$. Second, it adds finite off-isolation leak-path cases by forcing the router leak resistance down to $1\,\text{k}\Omega$. Third, it repairs the chatter case so that the handoff threshold and the collapse windows actually create a first handoff, a fall event, and a second rise in the vulnerable designs. Those changes matter because a falsifier that never induces the advertised failure mode cannot support a mechanism claim.

4.3 Robustness Protocol

The robustness study evaluates the four load-bearing cases `fa_001`, `fa_004`, `fa_005`, and `sm_015`. For each case-design pair, the script draws 24 independent samples with seed

$$s = 20260312 + 10000\,o_c + 1000\,o_d + n, \tag{6}$$

where o_c is the case offset, o_d the design offset, and $n \in \{0, \ldots, 23\}$ the sample index. The randomized parameters are:

- source resistances: multiplicative Gaussian variation with standard deviation 8% and floor factor 0.6;
- store capacitance: multiplicative Gaussian variation with standard deviation 10% and floor factor 0.5;
- startup dead-zone: multiplicative Gaussian variation with standard deviation 12%, clipped to 1 mV to 15 mV;
- RC-ranked and blind packet controls: scout R, scout C, and control conductance with standard deviations 12%, 12%, and 10%;
- time-constant-ranked control: fast/slow R and C values with standard deviation 12%, plus 10% control-conductance variation;
- leak resistance: 15% multiplicative Gaussian variation in the leak-path cases;
- collapse timing: additive Gaussian jitter with standard deviation 0.12 s;
- collapse scales: additive Gaussian jitter with standard deviation 0.04, clipped to $[0, 1]$.

Wilson 95% intervals are reported for startup probability. Since the study is simulation-only, those intervals quantify only the scripted parameter uncertainty, not fabrication variation or model inadequacy.

4.4 Software and Compute Environment

The experiments were executed on a Linux x86-64 workstation with 20 logical CPUs, Python 3.10.17, and `ngspice-39`. The artifact bundle, netlists, figures, and manifests are stored in the repository snapshot cited as [4]. All deterministic and robustness scripts are included in the same workspace.

5 Theoretical Results and Proofs

This section proves three statements directly from the implemented netlists. They are intentionally modest. The paper does not attempt a transistor-level proof of startup probability. It proves the structural facts that are actually available from the model and then tests the remaining claims experimentally.

Proposition 1 (Symmetry reduction of RC-Rankedto Blind Packet). *Assume the two routed branch voltages satisfy*

$$u_A(t) = u_B(t) \qquad \textit{for all } t \in [0, T], \tag{7}$$

and that the scout networks in the RC-ranked gate use identical component values and zero initial charge. Then, on $[0, T]$, the RC-ranked packet gate is dynamically identical to the blind packet gate: its branch weights satisfy $w_A(t) = w_B(t) = \frac{1}{2}$, and its pump and storage current equations reduce exactly to the blind-packet equations.

Proof. Let $q_A(t)$ and $q_B(t)$ denote the scout capacitor voltages `a_probe` and `b_probe`. Because the two probe branches use the same `SCOUT_R` and `SCOUT_C` values, each obeys the same first-order linear differential equation with the corresponding routed source voltage as input:

$$R_{\text{scout}}C_{\text{scout}}\,\dot{q}_A(t) + q_A(t) = u_A(t), \tag{8}$$

$$R_{\text{scout}}C_{\text{scout}}\,\dot{q}_B(t) + q_B(t) = u_B(t). \tag{9}$$

Subtract the second equation from the first and define $z(t) = q_A(t) - q_B(t)$. Then

$$R_{\text{scout}}C_{\text{scout}}\,\dot{z}(t) + z(t) = u_A(t) - u_B(t). \tag{10}$$

By assumption, the right-hand side is identically zero on $[0, T]$. The scout capacitors are initialized equally at zero, so $z(0) = 0$. The homogeneous linear differential equation with zero initial condition has the unique solution $z(t) = 0$ for all $t \in [0, T]$. Therefore $q_A(t) = q_B(t)$ for all $t \in [0, T]$.

The RC-ranked selector law is

$$w_A(t) = \frac{1 + \tanh\big(k_{\text{sel}}(q_A(t) - q_B(t))\big)}{2}, \qquad w_B(t) = 1 - w_A(t). \tag{11}$$

Since $q_A(t) - q_B(t) = 0$, the argument of the hyperbolic tangent is zero. Because $\tanh(0) = 0$, we obtain

$$w_A(t) = \frac{1+0}{2} = \frac{1}{2}, \qquad w_B(t) = 1 - \frac{1}{2} = \frac{1}{2}. \tag{12}$$

Substituting these weights into the RC-ranked branch current expressions gives

$$i_{\text{store}}(t) = \eta_{\text{pump}}G_{\text{pump}}\left(x_A(t)\frac{1}{2} + x_B(t)\frac{1}{2}\right), \tag{13}$$

$$i_{A,\text{in}}(t) = G_{\text{pump}}\,x_A(t)\frac{1}{2}, \qquad i_{B,\text{in}}(t) = G_{\text{pump}}\,x_B(t)\frac{1}{2}, \tag{14}$$

which are exactly the blind-packet equations. The control branch and the handoff latch are already shared between the two designs, so the full netlist dynamics coincide on $[0, T]$. □

Proposition 1 does not imply that the two packet-gated designs are always identical, because many cases intentionally violate branch symmetry. It does show, however, that explicit ranking cannot create a difference in any operating regime where the routed branch histories are already matched.

Theorem 1 (Exact pre-handoff control-energy law)**.** *For each implemented design d, the repaired pre-handoff control energy is exactly*

$$E^{(d)}_{\text{ctrl}} = G^{(d)}_{\text{ctrl}} \int_0^{\tau_d} v_{\text{store}}(t)^2\,dt, \tag{15}$$

where τ_d is the first handoff-latch time (or T_{stop} if no handoff occurs) and the effective control conductance is

$$G^{(\textsc{Fixed})}_{\text{ctrl}} = \frac{1}{40\,\text{M}\Omega} + \frac{1}{80\,\text{M}\Omega} = 37.5\,\text{nS}, \tag{16}$$

$$G^{(\textsc{Nonaware})}_{\text{ctrl}} = \frac{1}{40\,\text{M}\Omega} + \frac{1}{80\,\text{M}\Omega} + \frac{1}{80\,\text{M}\Omega} = 50\,\text{nS}, \tag{17}$$

$$G^{(RC\text{-}\textsc{Ranked})}_{\text{ctrl}} = G^{(\textsc{Blind Packet})}_{\text{ctrl}} = 40\,\text{nS}, \tag{18}$$

$$G^{(TC\text{-}\textsc{Ranked})}_{\text{ctrl}} = 18\,\text{nS}. \tag{19}$$

Proof. We proceed design by design. In every design, the control-monitor element `VCTRL_MON` is an ideal zero-volt source from `n_store` to `n_ctrl`. Therefore the source enforces

$$V(n_{\text{ctrl}}(t)) = V(n_{\text{store}}(t)) = v_{\text{store}}(t) \tag{20}$$

for all t in the transient analysis.

The measurement hook defines

$$p_{\text{ctrl}}(t) = \max\{v_{\text{store}}(t)\, I(\texttt{VCTRL\_MON})(t), 0\} \tag{21}$$

and then multiplies it by the pre-handoff factor $1 - \ell_d(t)$, where $\ell_d(t) \in [0, 1]$ is the nondecreasing handoff latch. Since every control branch draws current from `n_ctrl` to ground and $v_{\text{store}}(t) \geq 0$ throughout the startup interval, the current through `VCTRL_MON` is nonnegative whenever the control network is active. The outer maximum is therefore inactive, so

$$p_{\text{ctrl}}(t) = v_{\text{store}}(t)\, I(\texttt{VCTRL\_MON})(t). \tag{22}$$

For the fixed design, the control network consists of two resistors from `n_ctrl` to ground: `RCTRL_OSC` $= 40\,\text{M}\Omega$ and `RCTRL_PATH` $= 80\,\text{M}\Omega$. By Ohm's law and (20), the current through the monitor source is

$$I(\texttt{VCTRL\_MON})(t) = \frac{v_{\text{store}}(t)}{40\,\text{M}\Omega} + \frac{v_{\text{store}}(t)}{80\,\text{M}\Omega} = G_{\text{ctrl}}^{(\textsc{Fixed})}\, v_{\text{store}}(t). \tag{23}$$

Substitute this into (22) to obtain

$$p_{\text{ctrl}}^{(\textsc{Fixed})}(t) = G_{\text{ctrl}}^{(\textsc{Fixed})}\, v_{\text{store}}(t)^2. \tag{24}$$

Integrating only while the handoff latch is open yields

$$E_{\text{ctrl}}^{(\textsc{Fixed})} = \int_0^{T_{\text{stop}}} p_{\text{ctrl}}^{(\textsc{Fixed})}(t)(1 - \ell_{\text{fix}}(t))\, dt = G_{\text{ctrl}}^{(\textsc{Fixed})} \int_0^{\tau_{\text{fix}}} v_{\text{store}}(t)^2\, dt. \tag{25}$$

The nonaware design adds one extra $80\,\text{M}\Omega$ branch, so the monitor current becomes

$$I(\texttt{VCTRL\_MON})(t) = \left(\frac{1}{40\,\text{M}\Omega} + \frac{1}{80\,\text{M}\Omega} + \frac{1}{80\,\text{M}\Omega}\right) v_{\text{store}}(t) = G_{\text{ctrl}}^{(\textsc{Nonaware})} v_{\text{store}}(t), \tag{26}$$

and the same integration step gives the claimed expression for $E_{\text{ctrl}}^{(\textsc{Nonaware})}$.

The RC-ranked and blind packet-gated designs use a single behavioral control sink of the form

$$I_{\text{ctrl}}(t) = G_{\text{scout}}\, V(n_{\text{ctrl}}(t)) = G_{\text{scout}}\, v_{\text{store}}(t) \tag{27}$$

with $G_{\text{scout}} = 40\,\text{nS}$. Therefore

$$p_{\text{ctrl}}^{(\text{RC-}\textsc{Ranked})}(t) = p_{\text{ctrl}}^{(\textsc{Blind Packet})}(t) = G_{\text{scout}}\, v_{\text{store}}(t)^2, \tag{28}$$

and integrating to the respective handoff latch times yields the stated formulas.

The time-constant-ranked design has the same structure, but its control sink is `BCTRL_TC` with conductance $G_{\text{TC}} = 18\,\text{nS}$. Hence

$$p_{\text{ctrl}}^{(\text{TC-}\textsc{Ranked})}(t) = G_{\text{TC}}\, v_{\text{store}}(t)^2, \tag{29}$$

which integrates to the claimed result.

All cases therefore share the same generic law

$$E_{\mathrm{ctrl}}^{(d)} = G_{\mathrm{ctrl}}^{(d)} \int_0^{\tau_d} v_{\mathrm{store}}(t)^2 \, dt. \tag{30}$$

This completes the proof. □

Corollary 1 (Matched-trajectory energy ratio). *If two designs d_1 and d_2 satisfy $v_{\mathrm{store}}^{(d_1)}(t) = v_{\mathrm{store}}^{(d_2)}(t)$ for all $t \in [0, \tau]$ and hand off at the same time τ, then*

$$\frac{E_{\mathrm{ctrl}}^{(d_1)}}{E_{\mathrm{ctrl}}^{(d_2)}} = \frac{G_{\mathrm{ctrl}}^{(d_1)}}{G_{\mathrm{ctrl}}^{(d_2)}}. \tag{31}$$

*In particular, a matched TC-*Ranked*trajectory should consume only $18/40 = 0.45$ of the RC-*Ranked*pre-handoff control energy.*

Proof. Under the stated assumptions, the two integrals $\int_0^\tau v_{\mathrm{store}}(t)^2 dt$ are identical. Dividing the expressions from Theorem 1 immediately yields the conductance ratio. Substituting $G_{\mathrm{ctrl}}^{(\text{TC-Ranked})} = 18\,\mathrm{nS}$ and $G_{\mathrm{ctrl}}^{(\text{RC-Ranked})} = 40\,\mathrm{nS}$ gives $18/40 = 0.45$. □

Proposition 2 (Only NonawareContains an explicit pre-handoff branch-joining path). *Before handoff, the nonaware netlist contains finite-resistance paths from the routed branch-A bus to the routed branch-B bus, while the fixed and all three packet-gated netlists do not.*

Proof. We inspect each implementation directly.

Nonaware. The nonaware netlist includes

$$\texttt{RJOIN\_A\_P} \;\; n_{A,p} \leftrightarrow n_{\mathrm{start},p} \quad (1\,\mathrm{m\Omega}), \tag{32}$$
$$\texttt{RJOIN\_B\_P} \;\; n_{B,p} \leftrightarrow n_{\mathrm{start},p} \quad (1\,\mathrm{m\Omega}), \tag{33}$$
$$\texttt{RJOIN\_A\_N} \;\; n_{A,n} \leftrightarrow n_{\mathrm{start},n} \quad (1\,\mathrm{m\Omega}), \tag{34}$$
$$\texttt{RJOIN\_B\_N} \;\; n_{B,n} \leftrightarrow n_{\mathrm{start},n} \quad (1\,\mathrm{m\Omega}). \tag{35}$$

Therefore the positive buses are connected by the explicit resistor chain

$$n_{A,p} \longleftrightarrow n_{\mathrm{start},p} \longleftrightarrow n_{B,p} \tag{36}$$

with total resistance $2\,\mathrm{m\Omega}$, and the negative buses are connected by the analogous $2\,\mathrm{m\Omega}$ chain. These are finite, direct, pre-handoff branch-joining paths.

Fixed. The fixed design routes only source A into the startup pump. Source B is present in the source model and in the measurement hooks, but there is no resistor, switch, or pump input connecting the branch-B routed bus to the startup path. Hence there is no explicit pre-handoff branch-A to branch-B conductive path.

Packet-gated family. In the RC-ranked, blind, and time-constant-ranked netlists, the routed buses of A and B appear only as input arguments to separate behavioral current sources. Each branch injects current into the common store node through its own current source expression, but there is no resistor, switch, or explicit bus-join element connecting branch A to branch B. A current source from branch A to the store node and a second current source from branch B to the same store node do not create a finite-resistance branch-to-branch path, because the constitutive law of each source depends on its local terminal voltage and not on a direct conductive link to the other branch bus.

Consequently, before handoff, only NONAWAREcontains an explicit finite branch-joining path between the routed inputs. The fixed and packet-gated designs do not. □

Proposition 2 does not prove that NONAWAREmust fail every adversarial case. It proves the structural asymmetry that makes NONAWAREuniquely vulnerable to branch-loading and leak-path stress. The experimental sections test whether that structural difference is large enough to matter in the chosen weak-source regime.

6 Experimental Setup

6.1 Primary Startup Matrix

The deterministic startup matrix spans four nominal open-circuit voltages $\{20\,\mathrm{mV}, 50\,\mathrm{mV}, 100\,\mathrm{mV}, 300\,\mathrm{mV}\}$, four ramp rates $\{0.1\,\mathrm{mV\,s^{-1}}, 1\,\mathrm{mV\,s^{-1}}, 10\,\mathrm{mV\,s^{-1}}, 100\,\mathrm{mV\,s^{-1}}\}$, three impedance ratios $\{1, 5, 20\}$, and two polarity modes (same and mixed). The executed matrix contains 24 cases rather than the full 96-point cross, so each level is represented but not independently crossed inside every block. That choice keeps the runtime within the lane budget but limits causal interpretation.

All primary cases use $C_{\text{store}} = 10\,\mu\mathrm{F}$, $R_{\text{load}} = 100\,\mathrm{M\Omega}$, $R_A = 1\,\mathrm{k\Omega}$, and $R_B \in \{1\,\mathrm{k\Omega}, 5\,\mathrm{k\Omega}, 20\,\mathrm{k\Omega}\}$. Same-polarity cases use $(p_A, p_B) = (+1, +1)$; mixed-polarity cases use $(+1, -1)$.

6.2 Expanded Falsifier Suite

The falsifier suite contains ten cases.

- `fa_001`: same-polarity source collapse with moderate impedance asymmetry.
- `fa_002`: mixed-polarity source collapse with ratio 1:20.
- `fa_003`: mixed-polarity low-voltage asymmetry without explicit collapse.
- `fa_004`: real UVLO chatter with lowered handoff thresholds and delayed dual collapse.
- `fa_005`: mixed-polarity conflict with a heavy load and collapse of one source.
- `fa_006`: dual collapse under mixed polarity and heavy load.
- `fa_007` and `fa_008`: unequal open-circuit voltage cases in same and mixed polarity.
- `fa_009` and `fa_010`: same and mixed-polarity leak-path cases with router leak resistance forced to $1\,\mathrm{k\Omega}$.

Appendix B gives the complete numeric manifest.

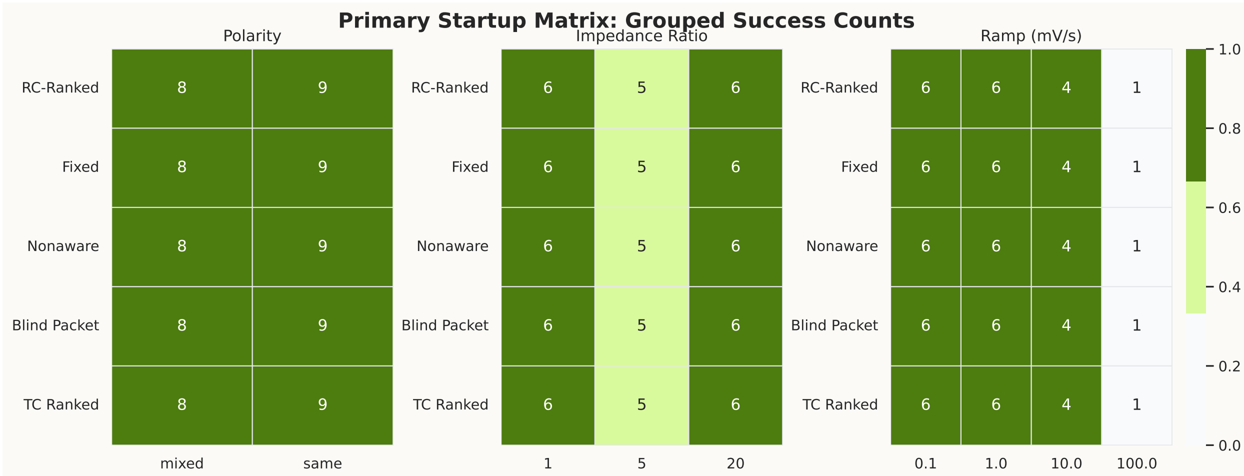


Figure 3: Grouped success counts for the 24-case primary startup matrix. Each row corresponds to one design and each column group to one factor family. Every row is identical: the entire five-design pack is tied at 17/24 startups. This figure is intentionally a null result. It shows that the global startup envelope in the chosen matrix is unchanged by adding packet gating, by joining both inputs, or by changing the pre-handoff selector law.

6.3 Result Aggregation and Confidence Intervals

The primary matrix and falsifier suite are deterministic. Their counts are exact within the simulated model and are therefore reported as counts rather than as estimated means. The robustness study adds Wilson 95% intervals on startup probability for the selected cases. Because all 24-sample runs in a success-only group have identical observed probability, the lower bound for those groups is 0.8620 rather than 1.0, reflecting finite sample size.

7 Results

7.1 Primary Startup Matrix: a Null Result

Figure 3 is the first major result, and it is a null. Every row is identical. All five designs start in 17/24 cases. All five achieve the same grouped counts: 8 mixed-polarity successes and 9 same-polarity successes; ratio-group counts of 6, 5, and 6; and ramp-group counts of 6, 6, 4, and 1 for $0.1\,\mathrm{mV\,s^{-1}}$, $1\,\mathrm{mV\,s^{-1}}$, $10\,\mathrm{mV\,s^{-1}}$ and $100\,\mathrm{mV\,s^{-1}}$. The startup envelope in the chosen 24-case matrix is therefore dominated by the source model and the load/storage contract, not by the selector law.

Table 3 quantifies the tie. The median successful-case handoff times are similar, ranging from $18.212\,4\,\mathrm{s}$ to $18.563\,7\,\mathrm{s}$. Even the source-blind and time-constant-ranked variants do not change the startup-matrix success count. That null result matters because it kills any honest attempt to sell the paper as a broad improvement in startup envelope.

7.2 Expanded Falsifiers Overturn the Original Hypothesis

Figure 4 is the paper's central experimental figure. It overturns the original RC-ranked story in two independent ways.

Design	Startup matrix successes	Falsifier successes	Median successful t_{handoff} (s)	Median successful E_{ctrl} (J)
RC-Ranked	17/24	8/10	18.5461	2.37717×10^{-13}
Fixed	17/24	7/10	18.2124	2.13446×10^{-13}
Nonaware	17/24	5/10	18.2711	2.91240×10^{-13}
Blind Packet	17/24	10/10	18.5637	2.37717×10^{-13}
TC-Ranked	17/24	9/10	18.3911	1.11852×10^{-13}

Table 3: Headline quantitative summary from the deterministic evidence pack. The startup-matrix tie is exact. The expanded falsifier suite overturns the original mechanism story by placing the blind packet gate above the RC-ranked design.

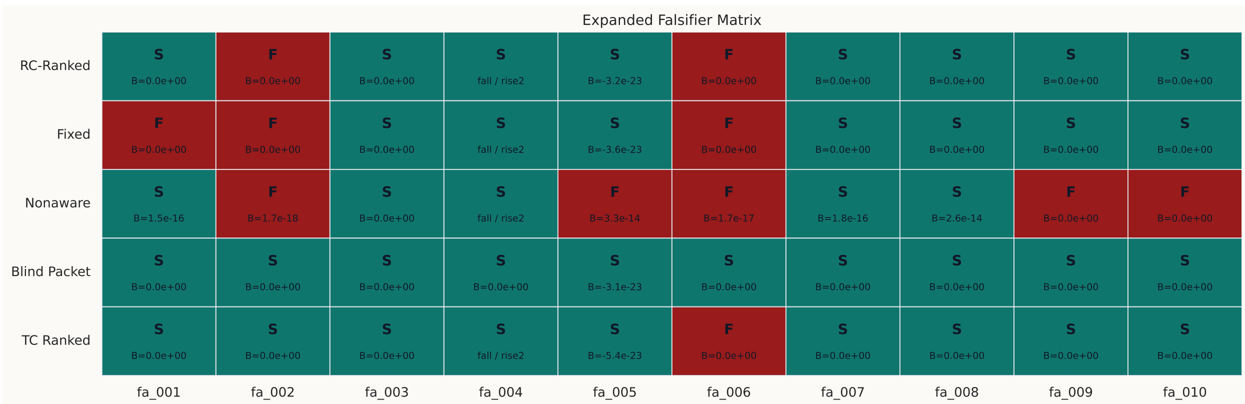


Figure 4: Expanded 10-case falsifier matrix. Green cells denote successful startup; red cells denote failure. Text within each cell reports measured pre-handoff back-drive energy or chatter events. The important pattern is row-wise, not column-wise: the blind packet gate is the only design that reaches 10/10 and the only one that avoids the measured fall/rise2 chatter event in `fa_004`. The nonaware design fails both leak-path cases, the mixed-conflict case, and remains weaker on collapse, consistent with its explicit pre-handoff branch join.

First, Blind Packetreaches 10/10 deterministic startups, while RC-Rankedreaches only 8/10. The only cases that separate those two designs are `fa_002` and `fa_006`, both of which are won by the blind packet gate. This means the executed same-family ablation does more than "match" the champion. It dominates it.

Second, the failure localization lines up with Proposition 2. The Nonawaredesign uniquely fails both leak-path cases (`fa_009` and `fa_010`) and the mixed-conflict case `fa_005`. The fixed design avoids the explicit join failure but still fails the same-polarity collapse case `fa_001`. Packet-gated isolation therefore explains the surviving boundary more parsimoniously than explicit ranking does.

Two case studies are especially informative.

- **`fa_001`.** The fixed design fails deterministically, Nonawarestarts late at 10.935 9 s with nonzero back-drive, and the three packet-gated designs start near 5.2 s–5.4 s with zero back-drive. This is the cleanest fixed-baseline separator.
- **`fa_004`.** The repaired chatter case finally measures what it was supposed to measure. RC-Ranked, Fixed, Nonaware, and TC-Rankedall show a first handoff near 7 s, a fall near 11.2 s, and a second rise near 15.9 s. Blind Packetstarts earlier at 3.837 39 s and never falls

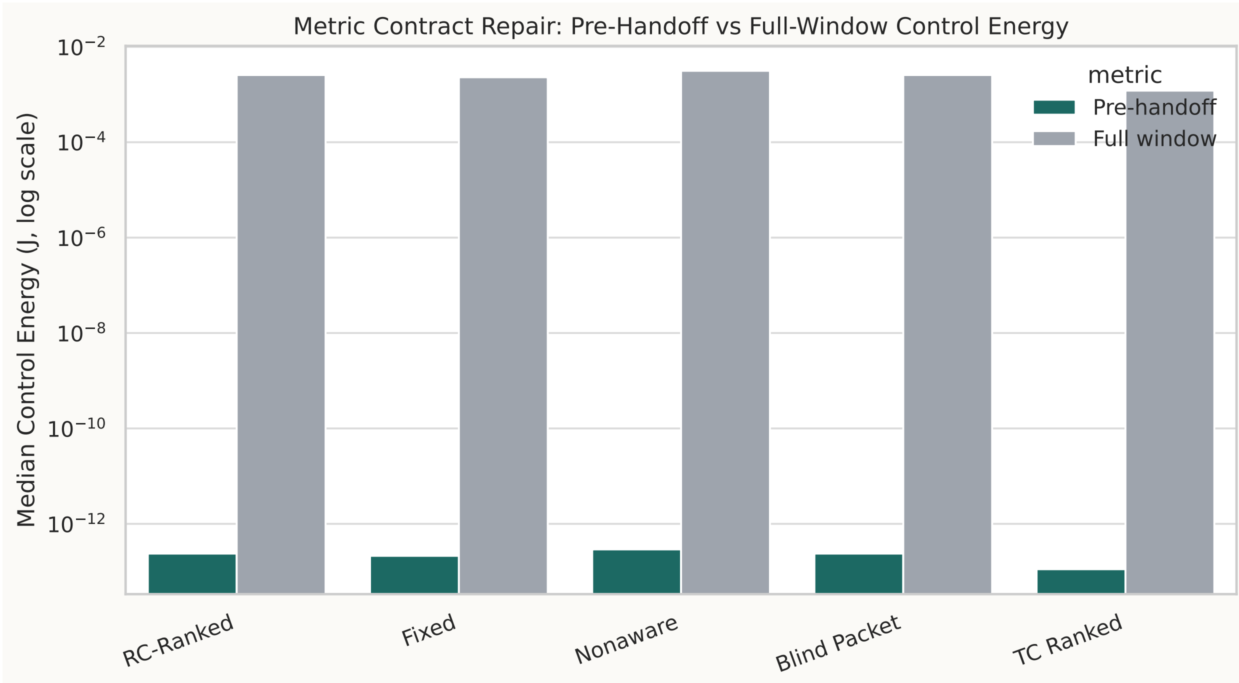


Figure 5: Median control energy under the repaired pre-handoff contract and the legacy full-window diagnostic. The full-window integral exaggerates every design by about 1.08×10^{10} because it continues to count post-handoff activity. The repaired metric exposes the actual pre-handoff ordering: RC-RANKEDand BLIND PACKETare equal on the startup matrix, NONAWAREis the largest, and TC-RANKEDis the smallest. Theorem 1 explains this ordering through effective control conductance.

back through the threshold. The blind packet gate therefore does not merely preserve startup; it removes the measured chatter event within this model.

7.3 Corrected Metric Accounting Restores Interpretability

The repaired measurement hook is not cosmetic. Figure 5 shows that full-window control-energy integration inflates all medians by a factor of approximately 1.08×10^{10} and largely obscures the distinctions that matter before handoff. Under corrected pre-handoff accounting, RC-RANKEDand BLIND PACKEThave the same successful-case median control energy, as predicted by Theorem 1, while TC-RANKEDis substantially lower.

Corollary 1 predicts that a matched TC-RANKEDtrajectory should consume only 45% of the RC-RANKEDcontrol energy. The measured successful-case median ratio is

$$\frac{1.11852 \times 10^{-13}}{2.37717 \times 10^{-13}} = 0.4705, \tag{37}$$

which is close enough to the ideal conductance ratio to support the claim that the energy reduction is mechanistic, not accidental. The remaining gap is explained by small differences in the actual store-voltage trajectories and handoff times.

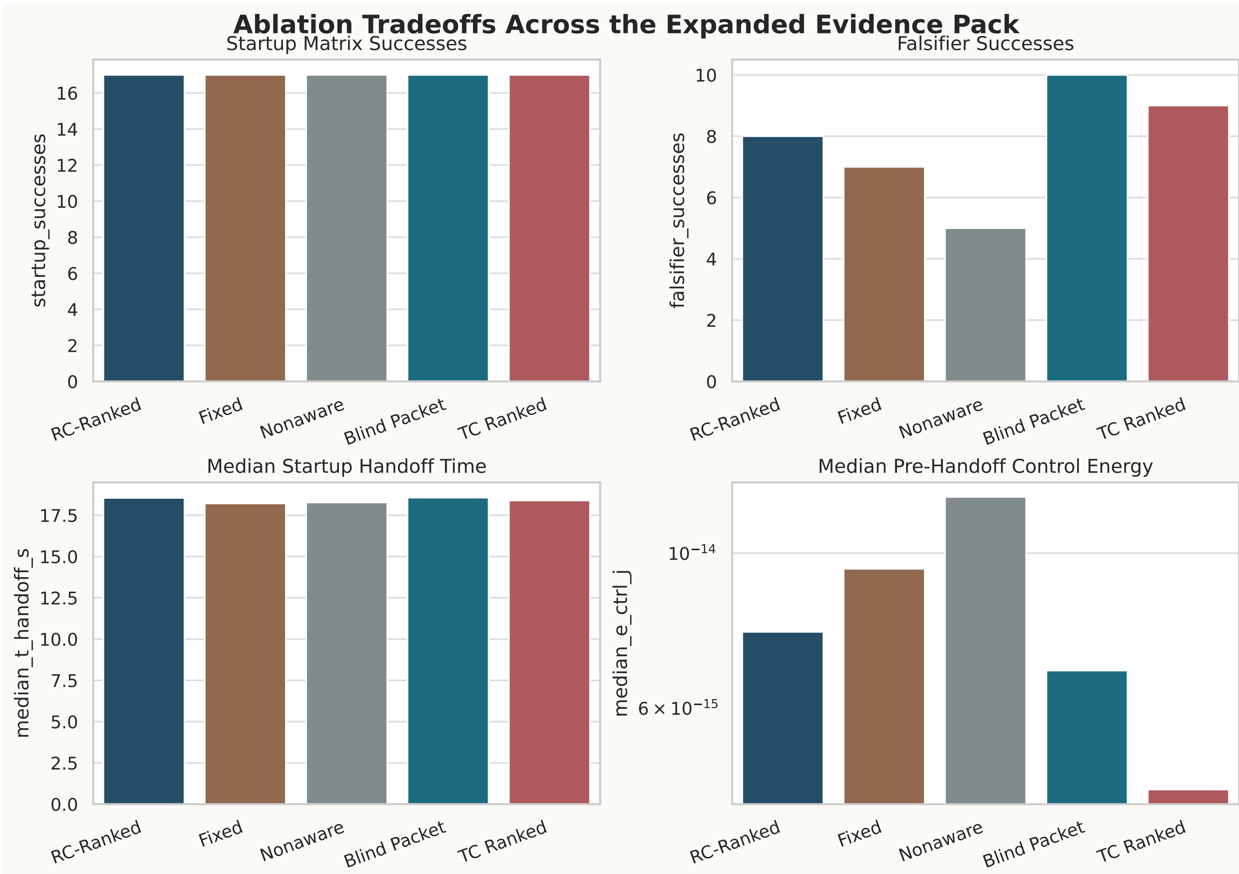


Figure 6: Ablation tradeoffs across the deterministic evidence pack. Startup-matrix success is identical for all five designs, so the global envelope is unchanged. The expanded falsifier suite then becomes the deciding discriminator, where BLIND PACKETis best and TC-RANKEDis second. The control-energy panel shows why TC-RANKEDremains valuable as a secondary result: it achieves nearly the same adversarial boundary with the lowest pre-handoff control cost.

7.4 Ablation Tradeoffs and Design Guidance

Figure 6 compresses the entire evidence pack into four panels. The top-left panel shows the startup-matrix tie. The top-right panel shows why the paper exists: the falsifier ranking is reversed relative to the original hypothesis. The bottom-left panel shows that handoff-time differences inside the startup matrix are minor. The bottom-right panel shows the engineering compromise. If one insists on keeping explicit source awareness inside the packet-gated family, then TC-RANKEDis the only defensible version. If one cares most about adversarial correctness in the executed model, BLIND PACKETis the dominant design.

7.5 Robustness: the Boundary Survives Randomized Perturbation

Figure 7 and Table 4 show the uncertainty-qualified version of the load-bearing cases. All success-only groups have the same Wilson lower bound of 0.8620 because they are based on 24/24 successes. The informative cases are the failure groups.

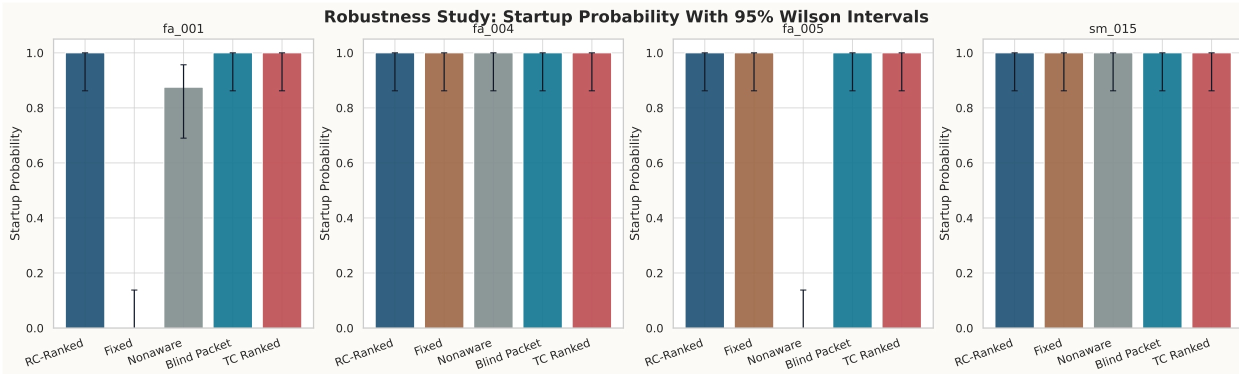


Figure 7: Startup probability with Wilson 95% intervals from 24 randomized runs per design-case. `fa_001` remains a clean separator against FIXED, and `fa_005` remains a clean separator against NONAWARE. `fa_004` and `sm_015` show that all five designs are robust in the chatter-repair case and in a representative primary-matrix null case. The packet-gated family therefore preserves its advantage under modest parameter jitter.

Case	Design	Startup rate	Wilson 95% interval	Median t_{handoff} on successful runs (s)
`fa_001`	FIXED	$0/24 = 0.000$	$[0.000, 0.138]$	—
`fa_001`	NONAWARE	$21/24 = 0.875$	$[0.690, 0.957]$	9.76317
`fa_001`	Packet-gated family	$24/24 = 1.000$	$[0.862, 1.000]$	5.11–5.41
`fa_005`	NONAWARE	$0/24 = 0.000$	$[0.000, 0.138]$	—
`fa_005`	Other four designs	$24/24 = 1.000$	$[0.862, 1.000]$	4.71–18.05
`sm_015`	All five designs	$24/24 = 1.000$	$[0.862, 1.000]$	17.89–19.92

Table 4: Selected robustness summary. Grouped rows are used when multiple designs share identical startup probability. The deterministic boundary remains visible after parameter jitter.

- In `fa_001`, the fixed design remains at 0/24 with a Wilson interval of $[0, 0.138]$, while NONAWAREremains at 21/24 with interval $[0.690, 0.957]$. The packet-gated designs remain 24/24.
- In `fa_005`, NONAWAREremains 0/24, while the other four designs remain 24/24.
- In `sm_015`, all five designs remain 24/24, which confirms that the primary-matrix null stays a null under the scripted perturbation.

8 Discussion

The paper's most important message is methodological. The same-family ablation should have been the first experiment, not the last. Once the blind packet gate is executed, the architectural story changes immediately. The correct conclusion is no longer "a smarter selector improves startup." The correct conclusion is "the selector was never the real ingredient; the isolation topology was." That is a stronger engineering insight precisely because it is negative.

The evidence also suggests a practical design rule. If the design goal is adversarial startup robustness inside this helper-free model, then a blind packet gate is sufficient and often better. If the design goal includes tighter control-energy budget without giving up most adversarial robustness, then a time-constant-ranked arbiter is the only source-aware variant that remains worth considering. In contrast, the original RC-ranked selector now appears to combine the cost of explicit ranking with none of the falsifier advantage that motivated it.

Several limitations matter.

- The primary startup matrix is aliased. It is useful for coverage but not for causal factor attribution.
- The study is simulation-only and uses behavioral startup pumps rather than a full transistor-level switched-converter realization.
- The literature comparison is structural, not measured. We did not implement a literature-faithful autonomous multi-input silicon comparator.
- The robustness study covers four load-bearing cases rather than the full deterministic manifest.

These limitations are serious enough to block a broad architecture paper, but they do not invalidate the narrower falsification result.

The societal implication is modest but real. Weak-source energy-harvesting interfaces are often discussed only through positive-result narratives: a new circuit, a new minimum startup voltage, a new efficiency peak. This paper argues for a more disciplined alternative. Publishing negative design-space results can reduce wasted complexity in batteryless sensing, especially when startup robustness matters more than nominal efficiency. The result is not that intelligence is useless. The result is that intelligence should earn its keep against the simplest nearby alternative, not against only the weakest baseline.

9 Conclusion

A helper-free RC-ranked packet gate does not win the weak multi-source cold-start problem in the tested model. The primary startup matrix is a five-way tie at 17/24 successes. The executed same-family ablation then overturns the original hypothesis: a blind packet gate reaches 10/10 startups on the expanded falsifier suite, while the RC-ranked design reaches only 8/10. A lower-overhead time-constant-ranked arbiter remains worthwhile as a secondary point, achieving 9/10 falsifier startups at 52.9% lower successful-case pre-handoff control energy than the RC-ranked design. The engineering conclusion is clear. Packet-gated isolation matters. Explicit pre-handoff source ranking does not.

A Extended Proof Details

A.1 Why the scout-state difference remains zero under symmetry

The proof of Proposition 1 relied on uniqueness of a first-order linear differential equation. For completeness, write the homogeneous solution explicitly:

$$z(t) = z(0) \exp\left(-\frac{t}{R_{\text{scout}} C_{\text{scout}}}\right). \tag{38}$$

Since $z(0) = 0$, the solution is identically zero. No hidden branch state can break that equality because the two scout capacitors are the only dynamic elements in the selector law.

A.2 Detailed conductance derivation

The conductance values in Table 2 follow directly from parallel combination.

$$G_{\mathrm{ctrl}}^{(\textsc{Fixed})} = \frac{1}{40\,\mathrm{M\Omega}} + \frac{1}{80\,\mathrm{M\Omega}} = 25\,\mathrm{nS} + 12.5\,\mathrm{nS} = 37.5\,\mathrm{nS}, \tag{39}$$

$$G_{\mathrm{ctrl}}^{(\textsc{Nonaware})} = 25\,\mathrm{nS} + 12.5\,\mathrm{nS} + 12.5\,\mathrm{nS} = 50\,\mathrm{nS}. \tag{40}$$

The packet-gated conductances are literal parameter values from the behavioral sinks. Thus the observed successful-case median ratio

$$\frac{E_{\mathrm{ctrl}}^{(\text{TC-}\textsc{Ranked})}}{E_{\mathrm{ctrl}}^{(\text{RC-}\textsc{Ranked})}} = \frac{1.11852 \times 10^{-13}}{2.37717 \times 10^{-13}} \approx 0.4705 \tag{41}$$

is close to the ideal conductance ratio $18/40 = 0.45$, while the observed ratio

$$\frac{E_{\mathrm{ctrl}}^{(\textsc{Nonaware})}}{E_{\mathrm{ctrl}}^{(\textsc{Fixed})}} \approx 1.364 \tag{42}$$

is close to the ideal ratio $50/37.5 = 1.333$. Those near-matches support the interpretation that pre-handoff control energy is being measured coherently after the metric repair.

B Detailed Case Manifest

Table 5 summarizes the expanded falsifier suite.

Case	Stress class	Polarity	Ratio	Key overrides
`fa_001`	source collapse	same	1:5	$R_{\mathrm{load}} = 1\,\mathrm{M\Omega}$, source A collapses to 5% at 4.2 s
`fa_002`	source collapse	mixed	1:20	$R_{\mathrm{load}} = 1\,\mathrm{M\Omega}$, source A collapses to 5% at 4.4 s
`fa_003`	polarity mismatch	mixed	1:20	no collapse; low-voltage mixed-polarity stress
`fa_004`	UVLO chatter	same	1:1	$V_{\mathrm{handoff}} = 0.8\,\mathrm{V}$, $V_{\mathrm{handoff,fall}} = 0.7\,\mathrm{V}$, delayed dual collapse, $R_{\mathrm{load}} = 100\,\mathrm{k\Omega}$
`fa_005`	mixed conflict	mixed	1:1	$R_{\mathrm{load}} = 50\,\mathrm{k\Omega}$, source A collapses fully at 4.6 s and recovers at 6.0 s
`fa_006`	dual collapse	mixed	1:20	heavy load, both sources collapse fully at different times

Case	Stress class	Polarity	Ratio	Key overrides
fa_007	unequal V_{OC}	same	1:5	$(V_A, V_B) = (20\,\mathrm{mV}, 50\,\mathrm{mV})$
fa_008	unequal V_{OC}	mixed	1:1	$(V_A, V_B) = (100\,\mathrm{mV}, 300\,\mathrm{mV})$
fa_009	leak path	same	1:5	router leak forced to $1\,\mathrm{k}\Omega$
fa_010	leak path	mixed	1:20	router leak forced to $1\,\mathrm{k}\Omega$

Table 5: Expanded falsifier manifest used in the paper. The suite broadens the original collapse-only package to include real chatter, unequal-source heterogeneity, and finite off-isolation leakage.

C Implementation and Reproduction Notes

The artifact bundle includes the shared netlist fragments, the deterministic manifests, the robustness script, and all generated figures [3, 4]. The paper-level regeneration sequence is:

1. run `python3 tools/run_h1_matrix.py run -designs champion,fixed,nonaware,source_blind,time_cons`
2. run `python3 tools/run_h1_falsifier.py run -designs champion,fixed,nonaware,source_blind,time_c`
3. run `python3 tools/run_h1_robustness.py -samples 24`;
4. run `python3 tools/analyze_h1_results.py`;
5. compile this manuscript with `pdflatex`, `bibtex`, and two final `pdflatex` passes.

All claim-bearing numeric values reported in the manuscript are taken from the generated tables rather than transcribed from raw logs.

References

[1] D. Alghisi, V. Ferrari, M. Ferrari, D. Crescini, F. Touati, and A. B. Mnaouer. Single- and multi-source battery-less power management circuits for piezoelectric energy harvesting systems. *Sensors and Actuators A: Physical*, 2017. doi: 10.1016/J.SNA.2017.07.027. URL https://www.semanticscholar.org/paper/9fd5e15b325f8534ee928242f7e9507a01586677.

[2] Mohammad Alhawari, B. Mohammad, H. Saleh, and Mohammed Ismail. An efficient polarity detection technique for thermoelectric harvester in l-based converters. *IEEE Transactions on Circuits and Systems Part I: Regular Papers*, 2017. doi: 10.1109/TCSI.2016.2619758. URL https://www.semanticscholar.org/paper/0f7b60950a42dd11c194be447eee4552488d6fc8.

[3] Archivara Repository Contributors. Experiment specification for the packet-scout h1 champion, 2026. Local repository document at `results/concept_evolve/tree/001_packet_scout_handoff_root/experiment_spec.md`.

[4] Archivara Repository Contributors. Archivara research workspace for the h1 multi-source cold-start study, 2026. Local repository consulted during the 2026-03-12 run.

[5] Peng Cao, Yao Qian, Pan Xue, Danzhu Lu, Jie He, and Zhiliang Hong. A bipolar-input thermoelectric energy-harvesting interface with boost/flyback hybrid converter and on-chip cold starter. *IEEE Journal of Solid-State Circuits*, 2019. doi: 10.1109/JSSC.2019.2924095. URL https://www.semanticscholar.org/paper/f2323b0968d9609749cf5fea6b1bcf46bd08cbd2.

[6] Zhidong Chen, Yinshui Xia, Ge Shi, et al. Self-powered collaborative energy harvesting interface circuit for stacked multiple piezoelectric elements. *IEEE Transactions on Power Electronics*, 2024. doi: 10.1109/TPEL.2024.3453921. URL https://www.semanticscholar.org/paper/0fd827923bda6369f44527bd52fdf7b4c60e4941.

[7] Mathieu Coustans, F. Krummenacher, and M. Kayal. A fully integrated 60 mv cold-start circuit for single coil dc-dc boost converter for thermoelectric energy harvesting. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 2019. doi: 10.1109/TCSII.2019.2922683. URL https://www.semanticscholar.org/paper/eaeaec370043b342b8dd9228ab57295db818ffa4.

[8] Abhik Das, Yuan Gao, and T. T. Kim. A 220-mv power-on-reset based self-starter with 2-nw quiescent power for thermoelectric energy harvesting systems. *IEEE Transactions on Circuits and Systems Part I: Regular Papers*, 2017. doi: 10.1109/TCSI.2016.2606122. URL https://www.semanticscholar.org/paper/d30bff950b51bfb17324047f1b9193d837341002.

[9] V. Gogolou, T. Noulis, and V. F. Pavlidis. Multi-source energy harvesting systems integrated in silicon: A comprehensive review. *Electronics*, 2025. doi: 10.3390/electronics14101951. URL https://www.semanticscholar.org/paper/6461dab6cdfcd007d17f484bb2f28744f7b08c1c.

[10] Jacob Göppert and Y. Manoli. Fully integrated startup at 70 mv of boost converters for thermoelectric energy harvesting. *IEEE Journal of Solid-State Circuits*, 2016. doi: 10.1109/JSSC.2016.2563782. URL https://www.semanticscholar.org/paper/c75c82be2e98a8d66907742a89b886902c1a0162.

[11] Qin Kuai, Ho-Yin Leung, Qiping Wan, and P. Mok. A high-efficiency dual-polarity thermoelectric energy-harvesting interface circuit with cold startup and fast-searching zcd. *IEEE Journal of Solid-State Circuits*, 2022. doi: 10.1109/JSSC.2021.3128625. URL https://www.semanticscholar.org/paper/8f9f44029e34598ce882795eacb740a3f1035fd3.

[12] Shuo Li, Xinjian Liu, and Benton Calhoun. A 32na fully autonomous multi-input single-inductor multi-output energy-harvesting and power-management platform with 1.2e5 dynamic range, integrated mppt, and multi-modal cold start-up. In *2022 IEEE International Solid-State Circuits Conference (ISSCC)*, 2022. doi: 10.1109/ISSCC42614.2022.9731732. URL https://www.semanticscholar.org/paper/44bcd7a3f69a9819a24e817d3f1ce2ea0fed3827.

[13] Roberto Rafael Flores Quintero and G. Espinosa Flores-Verdad. A cmos startup circuit for thermoelectric energy harvesting systems. *IEEE Latin America Transactions*, 2019. doi: 10.1109/TLA.2019.8826691. URL https://www.semanticscholar.org/paper/d9415d1253fb75da110cffc9d1ab87509010d30e.

[14] Yiwu Tang, C. Wong, Yuan Du, Li Du, Yilei Li, and Mau-Chung Frank Chang. A fully integrated 28nm cmos dual source adaptive thermoelectric and rf energy harvesting circuit with 110mv startup voltage. In *IEEE Custom Integrated Circuits Conference*, 2018. doi: 10.1109/CICC.2018.8357082. URL https://www.semanticscholar.org/paper/822e14d10b861cc415aa92b936edd5833b71cea4.

[15] Xiudeng Wang, Yinshui Xia, Zhangming Zhu, et al. Configurable hybrid energy synchronous extraction interface with serial stack resonance for multi-source energy harvesting. *IEEE Journal of Solid-State Circuits*, 2023. doi: 10.1109/JSSC.2022.3182118. URL https://www.semanticscholar.org/paper/254a50d3d11f95e13c87d5b068a92e29fb0f2dee.

[16] Zheng xin Weng, Yidie Ye, and Zhidong Chen. A self-powered multi-input osece interface circuit for multiple piezoelectric transducers. *IEICE Electronics Express*, 2024. doi: 10.1587/elex.21.20240504. URL https://www.semanticscholar.org/paper/6022e53796c25cba2130c9505c6e6c908fd43cb0.